

Multi-Asset Next-Day Return Prediction

Bitcoin, Gold and Silver — a leakage-audited, walk-forward-validated comparison of statistical, tree-based and recurrent models against the random-walk baseline, with a live prediction dashboard and API

Final Year Project Report

Bachelor of Science in Computer Science

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Repository layout, methodology, feature dictionary, results, limitations and viva notes are also provided as Markdown under docs/; every table in this report is generated from the files in results/ by docs/build_report.py.

Executive Summary

This project is a machine-learning system that forecasts the **next trading day's return** of three assets — Bitcoin (BTC-USD), Gold (GC=F futures) and Silver (SI=F futures) — and presents the forecast, together with its measured reliability, in a web dashboard and a REST API. The forecast target is the next-day log return $y_t = \ln(P_{t+1}/P_t)$; the price shown to the user is recovered as $P_t \cdot \exp(\hat{y})$. Daily data from Yahoo Finance (2018-01-01 to 2026-09-12; Bitcoin 3,147 usable days, Gold/Silver 2,156 exchange days) is enriched with macro series (US Dollar Index, WTI crude, 10-year Treasury yield, S&P 500, VIX) and the crypto Fear & Greed index.

Methodology. A frozen chronological split (train $\leq$ 2025-09-10, validation $\leq$ 2026-02-17, test = 2026-02-18 $\rightarrow$ 2026-09-12) with no shuffling; a scaler fitted on training rows only; 28–29 stationary, backward-looking features; hyper-parameters and the served model chosen by 4-fold expanding-window walk-forward validation inside train+val; and a single evaluation of the untouched test set. Ten model families are compared on identical days: a random-walk baseline, historical mean, ARIMA, Ridge, Random Forest, LightGBM, CatBoost, GRU, LSTM and a stacked ensemble. Every feature obeys a *close-time rule* — it may only use what is known when the asset's daily close is fixed (same-day macro data for Bitcoin, previous-day data for the metals, whose Yahoo close is the 13:30 ET COMEX settlement). Twelve specific data-leakage checks are enforced by a 34-case test suite.

Result. The validation-selected models are CatBoost for Bitcoin and Gold and a GRU for Silver. On the untouched test set all three assets sit at the random-walk floor: the served models' return-RMSE is 0.0–0.6 % *above* the naive forecast (Diebold–Mariano $p = 0.71$ – 0.99), directional accuracy 46.9 % / 51.7 % / 55.2 % (binomial $p = 0.83$ / 0.37 / 0.12) and R^2 in return space ≈ 0 . **No model beats the random walk at a one-day horizon for Bitcoin, Gold or Silver.** The result is consistent with weak-form market efficiency; the application reports the measured error next to every forecast. An intermediate version of the system reported 62 % directional accuracy for Silver; the final audit traced it to a close-time leak (same-day macro features against a 13:30 ET futures settlement), fixed it and re-ran the pipeline — the finding is reported in Section 9.

Improvement over the initial version. An audit of the original codebase found a price-reconstruction bug that corrupted every dollar metric, 30 % fabricated weekend rows for the metals, hyper-parameters selected on the test set, early stopping on training data, non-stationary price-level features, a served ensemble that predicted a constant, a broken inference path and misleading R^2 values (≈ 0.95 on price levels, which a random walk also achieves). All were fixed. A second, measured improvement phase then added $2.9\times$ more training data (2018 $\rightarrow$), volatility/momentum features, re-tuning, and logged design experiments; a final audit found and removed the close-time leak. On identical unseen days the served models moved by -0.6 % (Gold), -0.4 % (Silver) and $+0.1$ % (Bitcoin) in return-RMSE — inside noise: more data made the models equal to the random walk rather than slightly worse, not better than it.

Status. Working Streamlit dashboard (live forecast with uncertainty band, a *Predict-a-Day* demonstration on the unseen test period that reveals the actual price and the error, a full prediction history, validation/test tables, regime analysis and experiment tables) and FastAPI endpoints; reproducible pipeline (make pipeline); documentation for methodology, features, results, limitations, demonstration and viva.

1. Introduction

Short-horizon financial forecasting is a standard test case for machine learning because it is genuinely hard: daily returns of liquid assets are dominated by noise, exhibit fat tails and volatility clustering, and any predictable component is small and unstable. Many student and published projects nevertheless report near-perfect accuracy — almost always because the evaluation is flawed (price-level R^2 , random splits, test-set tuning, look-ahead in features). This project was built as a working prediction system for Bitcoin, Gold and Silver and then subjected to a full methodological audit. The final version answers a precise question — *does any of ten model families beat the random walk at a one-day horizon, and by how much?*

— with a pipeline in which every leakage path is closed and tested, and it serves the validation-selected model in a dashboard that always shows the forecast next to its held-out performance.

2. Project Objectives

- Build a reproducible, leakage-free pipeline from raw market data to next-day forecasts for three assets.
- Engineer stationary, financially motivated features from price, volume, macro and sentiment data.
- Compare baseline, classical, tree-based and deep models under expanding-window walk-forward validation, selecting hyper-parameters and the served model without touching the test set.
- Evaluate once on an untouched test period with metrics that are meaningful for returns (RMSE/MAE/R² in return space, directional accuracy with a significance test, Diebold–Mariano test against the random walk, strategy backtest).
- Deliver a demonstrable application (dashboard + API) that communicates the forecast, its horizon, the model used, its measured error and a clear disclaimer, and lets an evaluator predict any unseen day and compare with the actual value.
- Measure every design change (data size, features, horizon, target) on validation data before keeping it.
- Document the methodology, results, limitations and the audit-driven improvements for examination.

3. Project Scope

In scope: daily data; a one-trading-day horizon; Bitcoin (BTC-USD), Gold (GC=F front-month futures) and Silver (SI=F); free public data sources; ten model families; a Streamlit dashboard and a FastAPI service running locally or in Docker. Out of scope: intraday data, order-book or news-text features, multi-day horizons, portfolio construction, and any form of trading execution. The system is a research prototype and decision-support tool, not financial advice.

4. System Overview

The system is a single Python code base (Python 3.9, pandas, scikit-learn, LightGBM, CatBoost, TensorFlow/Keras 2.16, statsmodels, Streamlit 1.50, FastAPI) organised as a linear pipeline. Each stage is a script with a Make target; `make pipeline` reproduces every number in this report from the raw CSV files.



Figure 1. End-to-end workflow. The test split is read by exactly one script.

Stage	Module	What it does
Data collection	src/data/data_collection.py, external_data.py	Daily OHLCV for the three assets; DXY, WTI, ^TNX, ^GSPC, ^VIX; Fear & Greed index
Cleaning & features	src/data/preprocessing.py	De-duplication, trading-day calendar, log returns, indicators, ratio features, external alignment, frozen split, train-only scaler, build_dataset()
EDA	src/data/eda.py	ADF tests on price vs return, ACF/PACF, return distributions, volatility clustering
Models	src/models/registry.py, model_gru.py, model_lstm.py, ensemble_model.py	Uniform fit/predict interface for all models; two-phase fitting; stacked ensemble
Tuning	src/training/tune_models.py	4-fold expanding-window walk-forward grid search on train+val

Stage	Module	What it does
Training	src/training/train_models.py	Phase A (train → validation metrics, loss curves, importance) and phase B (refit on train+val)
Evaluation	src/evaluation/backtesting.py, cross_validation.py, plots.py	Single test-set evaluation, model selection, results tables, figures
Serving	src/inference/prediction.py, app/streamlit_app.py, src/api/app.py	Next-day forecast with context; dashboard; REST endpoints
Experiments	src/experiments/run_experiments.py, before_after.py	E1 data size, E2 feature ablation, E3 horizon, E4 volatility (validation only); before/after on identical unseen days
Quality	tests/ (34 cases), .github/workflows/ci.yml, Dockerfile	Look-ahead (crypto + futures), close-time alignment, split/embargo, reconstruction, metrics, demo-inference tests; CI; container

Table 1. Implemented components (source of truth: the repository).

5. Dataset and Data Processing

Series	Source / ticker	Rows used	Period	Role
Bitcoin OHLCV	Yahoo Finance, BTC-USD	3,147 calendar days	2018-01-31 → 2026-09-12	target + features
Gold futures OHLCV	Yahoo Finance, GC=F	2,156 exchange days	2018-02-15 → 2026-09-11	target + features
Silver futures OHLCV	Yahoo Finance, SI=F	2,156 exchange days	2018-02-15 → 2026-09-11	target + features
US Dollar Index, WTI crude, 10-y yield	Yahoo Finance: DX-Y.NYB, CL=F, ^TNX	aligned to asset calendar	2018-01 → 2026-09	metals features
S&P; 500, VIX	Yahoo Finance: ^GSPC, ^VIX	aligned	2018-01 → 2026-09	all assets
Crypto Fear & Greed index	alternative.me API	aligned	2018-02 → 2026-09	Bitcoin feature

Table 2. Data sources. Row counts are after indicator warm-up (30 rows). Raw downloads cover 2018-01-01 → 2026-09-12; the earlier 3-year download (2023-07 → 2026-07) is archived for the before/after comparison.

Cleaning (`DataCleaner.clean_data`): duplicate timestamps removed; rows sorted; non-positive prices dropped. Bitcoin is re-indexed to every calendar day (no gaps were present). Gold and Silver **keep their exchange calendar** — the original pipeline forward-filled weekends, which created ~30 % synthetic zero-return rows. External series are converted to daily log returns on their own calendar and aligned to the asset's calendar with a forward fill (only past values are carried forward). After feature computation the 30-row warm-up is dropped and the frame is validated (no NaN/inf).

Exploratory analysis (training window only) confirms the modelling choices: price levels are non-stationary, returns are stationary, fat-tailed and almost uncorrelated (Table 3, Figure 3). This is why the target is the return and why the random walk is a demanding baseline.

Asset	Days (total / train)	ADF p (price)	ADF p (return)	Daily σ %	Excess kurtosis	ACF lag-1
Bitcoin	3147 / 2780	0.97	< 1e-10	3.42	15.2	-0.053
Gold	2156 / 1904	1.00	< 1e-10	0.97	3.6	-0.027
Silver	2156 / 1904	0.97	< 1e-10	1.91	5.2	-0.021

Table 3. Stationarity and distribution statistics of daily log returns (results/eda_summary.csv).



Figure 2. Daily closes with the frozen train / validation / test windows.

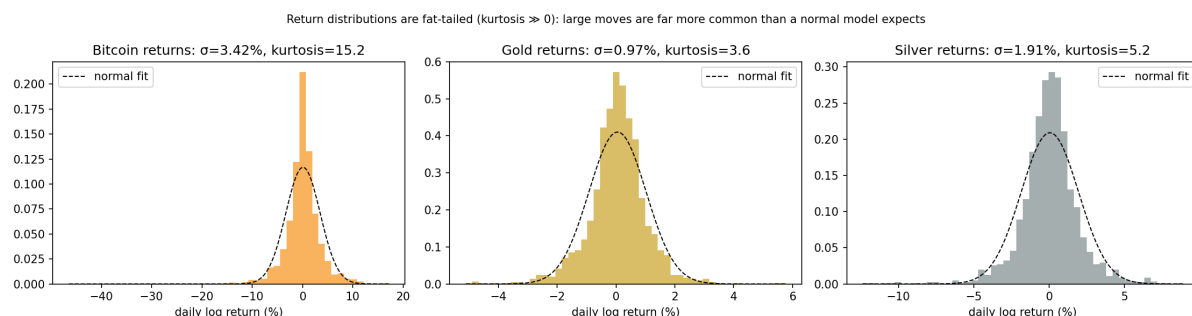


Figure 3. Return distributions are fat-tailed relative to a normal fit (training window).

6. Feature Engineering

All features at day t use only information known at the moment the day- t close is fixed — the 00:00 UTC bar close for Bitcoin, the 13:30 ET COMEX settlement for Gold and Silver. Because the settlement precedes the S&P 500 (16:00 ET), VIX (16:15), 10-year-yield (~15:00), DXY (17:00) and WTI (14:30) closes, and because Yahoo's futures High/Low span the whole Globex session, the metals use the *previous* day's macro returns and the previous session's High/Low-based features (hl_range, ATR/price, ADX); Bitcoin, whose bar closes after every US close, uses same-day values. Raw price levels (open, high, low, close, volume, EMA, Bollinger bands, ATR, MACD) are retained only for charts; the models receive scale-free ratios so that the feature distribution does not drift when the price trends outside the training range (Gold's test prices, \$3,986–5,294, lie entirely above its training range of \$1,817–3,644). A unit test perturbs the last observation and asserts that no earlier feature changes (no look-ahead).

Group	Features (formula)	Assets
Return path	$\log_return = \ln(P_t/P_{t-1})$; $return_1d/2d/5d/10d$ = lagged returns; $return_20d$ = 20-day cumulative return	all
Volatility (rolling / HAR / EWMA)	$volatility_10d/30d$ = rolling std; $rv_1d/5d/22d$ = $\sqrt{\text{mean } r^2 \text{ over } 1/5/22 \text{ days}}$; $ewma_vol$ = RiskMetrics σ ($\lambda = 0.94$)	all
Momentum / trend	RSI(14); ADX(14); ROC(12); $macd_norm = (EMA12-EMA26)/P$; $macd_hist_norm = (MACD-Signal9)/P$; $ema14_ratio$, $ema30_ratio = P/EMA-1$	all
Volatility state	$bb_pctb = (P-BB_low)/(BB_up-BB_low)$; $bb_width = (BB_up-BB_low)/BB_mid$; $atr_norm = ATR14/P$; $hl_range = (H-L)/P$	all
Macro	$sp500_return$, vix_return (all); dxy_return , oil_return , tnx_return (metals); $gold_return$ (silver). Previous-day values for the metals (close-time rule)	per asset
Sentiment / calendar / activity	$fear_greed$ (0–100); dow_sin , dow_cos ; $\log_volume_change$ (clipped ± 3)	Bitcoin

Table 4. Model input features: Bitcoin 29, Gold 28, Silver 29 (docs/FEATURES.md gives every formula and rationale). Groups added in the improvement phase are in the second row.

Deliberately excluded: futures volume for Gold/Silver (front-month contract volume from Yahoo jumps at contract rolls), day-of-week for the metals (no justification once synthetic weekends are removed), and redundant level indicators (VWAP, SMA, BB_Mid).

Target. $y_t = \ln(P_{t+1}/P_t)$, the next row's log return, created only inside `create_sequences`; the 30-day input window never contains that row (unit-tested). Recurrent models receive the $30 \times F$ window; tabular models receive the last row of the same window, so every model is scored on identical days. The target is standardised with the training mean and standard deviation and inverted exactly for reporting.

7. Machine Learning Models

Family	Model	Input	Notes
Baseline	Naive (random walk)	—	$\hat{y} = 0$, i.e. tomorrow's price = today's. The reference every model must beat.
Baseline	Historical mean	—	$\hat{y} = \text{mean training return (drift)}$.
Baseline	ARIMA(p,0,q)	return series	Order by AIC on train (BTC (1,0,0), Gold (2,0,2), Silver (3,0,2)); one-step walk-forward.
Linear	Ridge	last-day features	L2-regularised regression; α tuned.
Trees	Random Forest	last-day features	Depth-limited bagged trees.
Trees	LightGBM	last-day features	Gradient boosting, early-stopped on validation.
Trees	CatBoost	last-day features	Ordered boosting, early-stopped on validation. Served for Bitcoin and Gold.
Deep	GRU	$30 \times F$ window	One GRU layer (32 units) + dropout + dense; early stopping. Served for Silver.
Deep	LSTM	$30 \times F$ window	Same shape as the GRU for a fair comparison.
Ensemble	Stacked	OOF predictions	Non-negative Ridge over Ridge/LightGBM/CatBoost/GRU (experiment).

Table 5. Models compared (src/models/registry.py).

Two-phase training. Every model with a stopping rule (number of trees / boosting iterations / epochs) is first fitted on the training window while monitoring validation loss to find the stopping point (phase A — this also yields honest validation metrics and loss curves), then refitted on train+validation with that stopping point fixed (phase B — the deployed model). The validation data is therefore never inside its own early-stopping monitor.

Asset	Served model & tuned parameters	Stopping point	Other tuned configurations
Bitcoin	CatBoost: depth 7, lr 0.03, L2 3.0	50 rounds	Ridge α 100.0; RF depth 3; LightGBM 31 leaves; GRU 32u/drop 0.4; LSTM 32u/drop 0.4
Gold	CatBoost: depth 3, lr 0.01, L2 10.0	606 rounds	Ridge α 100.0; RF depth 3; LightGBM 31 leaves; GRU 32u/drop 0.4; LSTM 32u/drop 0.2
Silver	GRU: 32 units, dropout 0.4, lr 0.001, batch 64	9 epochs	Ridge α 100.0; RF depth 3; LightGBM 8 leaves; CatBoost depth 3; LSTM 32u/drop 0.2

Table 6. Hyper-parameters chosen by walk-forward validation (results/tuning/best_params.json). All tuned configurations of a model lie within 0.4–2 % of each other while the fold-to-fold standard deviation is ≈ 20 % of the mean; early stopping selected 10–82 boosting rounds for the metals' LightGBM/CatBoost (Gold CatBoost 606 at learning-rate 0.01) and 1–9 epochs for the recurrent models: the data supports very little capacity.

Which model is served, and why. Selection is automatic: the model with the lowest mean walk-forward RMSE of the return on the train+val folds — CatBoost for Bitcoin and Gold, a GRU for Silver. All nine models lie within 0.4–0.5 % of each other and within 0.35 % of the random walk on validation, so the ranking is not statistically decisive and is stated as such. The served Silver GRU illustrates the point: its predictions have a standard deviation of 0.0009 against a realised 0.032 — it is effectively a drift forecast that scored lowest on the folds. Boosted trees remain the practical choice (seconds to train, feature importance); the recurrent models stop after 1–9 epochs because validation loss never improves.

8. Training and Evaluation Methodology

Split	Rule	Bitcoin	Gold / Silver	Used for
Train	target $\leq$ 2025-09-10	2,750 windows (from 2018-03)	1,874 windows (from 2018-03)	fitting; scaler
Validation	2025-09-10 < target $\leq$ 2026-02-17	160 days	109 days	early stopping; walk-forward folds with train
Test	target > 2026-02-17 (to 2026-09-12)	207 days	143 days	evaluated once by backtesting.py

Table 7. Frozen chronological split (config.py), defined by the dates the target covers (exact embargo for multi-day targets). No shuffling; identical calendar boundaries for all assets.

Walk-forward validation. The train+validation period is cut into 4 consecutive validation blocks; for fold k the model is trained on everything before the block and scored on the block (expanding window). Inside each fold the last 15 % of the fold's training window is the early-stopping monitor. Hyper-parameter grids (over regularisation strength rather than capacity) and model selection use only these folds. k-fold cross-validation was rejected because it would train on 2026 to predict 2024 and leak regime information.

Metric	Space	Why it is reported
MAE, RMSE, MAPE	USD price	What a user experiences; MAPE is scale-free across a \$60,000 and a \$60 asset.
RMSE, MAE, R^2	log return	The quantity actually predicted; $R^2 < 0$ means worse than the mean. R^2 on price levels is NOT reported because a random walk scores ≈ 0.95 there.
Directional accuracy	sign	% of non-flat days with $\text{sign}(\hat{y}) = \text{sign}(y)$, with a one-sided binomial p-value against 50 %. A zero forecast has no direction ("—").
Diebold–Mariano test	log return	Formal test of equal squared-error accuracy against the random walk (HAC variance, small-sample correction).
Strategy backtest	—	Long/flat rule (long if $\hat{y} > 0$) with 10 bps cost vs buy-and-hold — economic relevance.
Train vs validation RMSE	log return	Over-fitting gap, compared with the random walk's own gap.

Table 8. Evaluation metrics (src/utls/metrics.py).

Leakage check	Mechanism	Verified by
Random shuffling	chronological date slicing	test_split_is_chronological_and_disjoint
Scaler fitted on val/test	MinMaxScaler.fit(train) only	test_scaler_fitted_on_train_only
Target inside input window	y = row i+30; window = rows i...i+29	test_create_sequences_target_is_next_row_and_never_in_window
Look-ahead in rolling / EWM / shift features	all operators backward-looking	test_no_lookahead_in_any_feature (crypto and futures paths)
External data closing after the asset (close-time rule)	same-day for BTC (00:00 UTC bar); latest value strictly before day t for the metals (13:30 ET settlement)	test_commodity_external_features_are_previous_day_values
Futures High/Low past the settlement	hl_range, atr_norm, ADX lagged one session for the metals	test_commodity_high_low_features_are_lagged_one_session
Early stopping on training data	two-phase fit	registry.py; CatBoost bug removed
Hyper-parameters chosen on test	tuner reads train+val arrays only	grep: no test reference in training/tuning/stacking
Synthetic rows	exchange calendar for futures	test_commodities_keep_trading_calendar
Wrong price reconstruction	explicit previous close	test_true_target_reconstructs_actual_close_exactly (error 1e-11)
Test set touched more than once	single script	backtesting.py is the only reader of the test split
Multi-day target straddling a split	split by target dates	test_task_targets_are_future_only_and_embargoed
Demo prediction seeing future rows	features.loc[:as_of] before scaling	test_predict_for_date... perturbs every future row; forecast unchanged

Table 9. Data-leakage checklist enforced by the test suite (tests/, 34 cases, all passing).

9. Experimental Results

Test period 2026-02-18 → 2026-09-12 for all assets (Bitcoin 207 days; Gold and Silver 143 exchange days). "CV" columns are the walk-forward validation scores used for selection; the remaining columns are the single test-set evaluation. The served model is marked "(served)". All values are from results/cv_results.csv and results/final_test_results.csv.

9.1 Bitcoin

Model	CV RMSE (ret) mean ± std	Test RMSE (ret)	Test R ² (ret)	MAE (\$)	RMSE (\$)	MAPE %	Dir. acc. % (p-value)	DM p vs naive
Naive (random walk)	0.03160 ± 0.00690	0.02079	-0.001	1,066.42	1,451.74	1.52	—	—
Historical mean	0.03163 ± 0.00694	0.02078	-0.000	1,067.32	1,451.38	1.53	49.3 (0.61)	0.81
ARIMA	0.03164 ± 0.00695	0.02091	-0.012	1,067.96	1,459.44	1.53	46.9 (0.83)	0.35
Ridge	0.03161 ± 0.00693	0.02080	-0.002	1,071.97	1,452.59	1.53	47.8 (0.76)	0.91
Random Forest	0.03160 ± 0.00688	0.02075	+0.004	1,064.48	1,448.81	1.52	48.3 (0.71)	0.62
LightGBM	0.03158 ± 0.00692	0.02068	+0.010	1,076.49	1,446.18	1.54	51.2 (0.39)	0.79
CatBoost (served)	0.03156 ± 0.00695	0.02082	-0.004	1,071.08	1,453.76	1.53	46.9 (0.83)	0.74
GRU	0.03168 ± 0.00697	0.02089	-0.010	1,075.86	1,458.31	1.54	46.9 (0.83)	0.35
LSTM	0.03170 ± 0.00697	0.02082	-0.004	1,068.06	1,453.44	1.53	46.4 (0.87)	0.35
Stacked ensemble	—	0.02065	+0.013	1,061.34	1,442.94	1.52	50.2 (0.50)	0.44

Table 10. Bitcoin: walk-forward validation and untouched test-set results, all models.

9.2 Gold

Model	CV RMSE (ret) mean $\pm$ std	Test RMSE (ret)	Test R ² (ret)	MAE (\$)	RMSE (\$)	MAPE %	Dir. acc. % (p-value)	DM p vs naive
Naive (random walk)	0.01114 $\pm$ 0.00225	0.01671	-0.002	58.26	75.76	1.29	—	—
Historical mean	0.01113 $\pm$ 0.00223	0.01675	-0.007	58.39	75.99	1.30	49.7 (0.57)	0.43
ARIMA	0.01123 $\pm$ 0.00219	0.01681	-0.015	58.57	76.27	1.30	50.3 (0.50)	0.52
Ridge	0.01112 $\pm$ 0.00223	0.01677	-0.009	58.51	76.05	1.30	50.3 (0.50)	0.76
Random Forest	0.01116 $\pm$ 0.00220	0.01659	+0.012	58.08	75.10	1.29	49.0 (0.63)	0.57
LightGBM	0.01113 $\pm$ 0.00223	0.01672	-0.003	58.25	75.81	1.29	49.7 (0.57)	0.88
CatBoost (served)	0.01110 $\pm$ 0.00219	0.01681	-0.015	58.43	76.22	1.30	51.7 (0.37)	0.71
GRU	0.01114 $\pm$ 0.00222	0.01664	+0.007	58.00	75.42	1.29	51.7 (0.37)	0.58
LSTM	0.01116 $\pm$ 0.00223	0.01687	-0.022	58.82	76.58	1.31	49.7 (0.57)	0.28
Stacked ensemble	—	0.01705	-0.043	58.95	77.28	1.31	51.0 (0.43)	0.44

Table 11. Gold: walk-forward validation and untouched test-set results, all models.

9.3 Silver

Model	CV RMSE (ret) mean $\pm$ std	Test RMSE (ret)	Test R ² (ret)	MAE (\$)	RMSE (\$)	MAPE %	Dir. acc. % (p-value)	DM p vs naive
Naive (random walk)	0.02391 $\pm$ 0.00624	0.03189	-0.001	1.79	2.37	2.49	—	—
Historical mean	0.02391 $\pm$ 0.00623	0.03192	-0.003	1.79	2.38	2.49	51.7 (0.37)	0.63
ARIMA	0.02410 $\pm$ 0.00613	0.03226	-0.024	1.81	2.37	2.52	52.4 (0.31)	0.50
Ridge	0.02393 $\pm$ 0.00625	0.03181	+0.005	1.79	2.37	2.48	54.5 (0.16)	0.48
Random Forest	0.02396 $\pm$ 0.00619	0.03146	+0.026	1.76	2.33	2.44	56.6 (0.07)	0.17
LightGBM	0.02389 $\pm$ 0.00625	0.03189	-0.001	1.79	2.37	2.48	58.7 (0.02)	0.98
CatBoost	0.02391 $\pm$ 0.00623	0.03193	-0.003	1.79	2.38	2.49	51.7 (0.37)	0.70
GRU (served)	0.02387 $\pm$ 0.00625	0.03189	-0.001	1.79	2.37	2.48	55.2 (0.12)	0.99
LSTM	0.02393 $\pm$ 0.00625	0.03198	-0.006	1.80	2.38	2.49	51.7 (0.37)	0.63
Stacked ensemble	—	0.03191	-0.002	1.79	2.38	2.48	52.4 (0.31)	0.91

Table 12. Silver: walk-forward validation and untouched test-set results, all models.

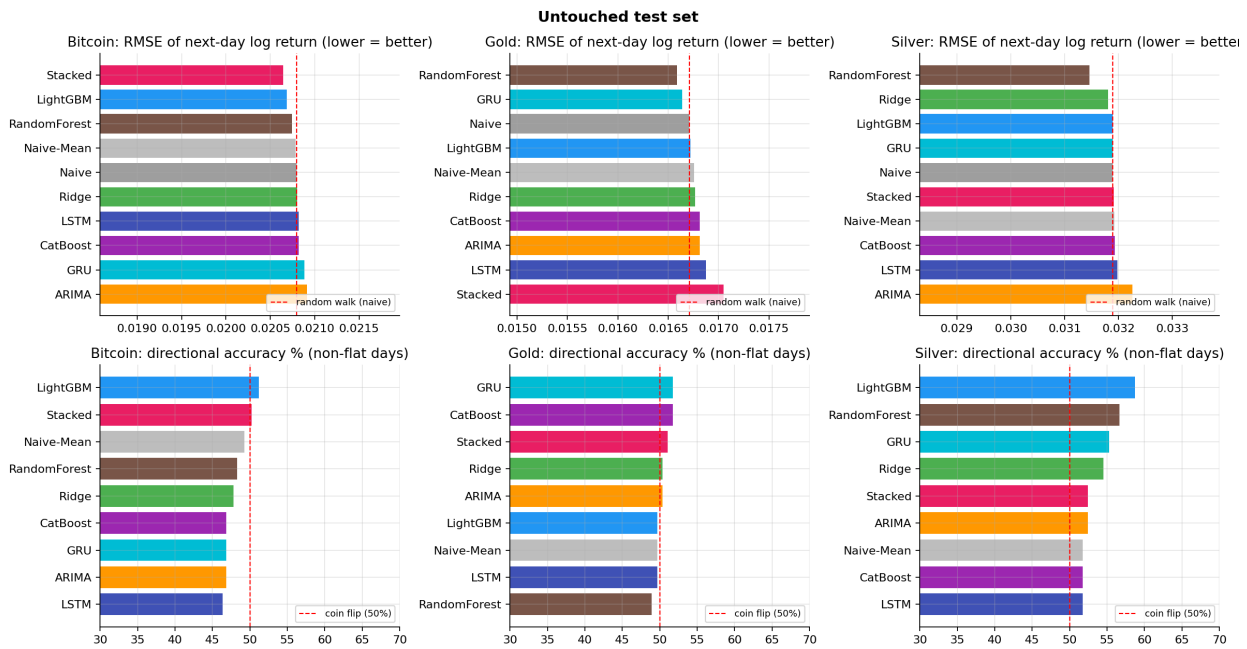


Figure 4. Test-set comparison: RMSE of the next-day return (red line = random walk) and directional accuracy (red line = 50 %).

9.4 Interpretation

- **All three assets are at the random-walk floor.** The served models' return-RMSE is 0.0–0.6 % above the naive forecast (Diebold–Mariano $p = 0.74$ Bitcoin, 0.71 Gold, 0.99 Silver); directional accuracy 46.9 % / 51.7 % / 55.2 % (binomial $p = 0.83$ / 0.37 / 0.12); R^2 in return space -0.004 / -0.015 / -0.001 . The tuned models are close to the unconditional drift, and the USD errors (MAPE 1.53 / 1.30 / 2.48 %) equal the random walk's (1.52 / 1.29 / 2.49 %) because they measure the asset's volatility, not skill.
- **The one $p < 0.05$ cell.** Silver LightGBM scores 58.7 % directional accuracy ($p = 0.02$) but has exactly the random walk's RMSE (DM $p = 0.98$) and a walk-forward figure of 53.3 %; with 27 model-rows in the test table one such cell is what chance produces (Bonferroni threshold ≈ 0.002). It is not the validation winner and is not claimed.
- **The leak that was found and removed (Section 9.6).** An intermediate version of this system aligned the metals' macro features same-day and reported 62.2 % directional accuracy for Silver (DM $p = 0.002$); Yahoo's futures close is the 13:30 ET settlement, so those features contained 1–3.5 hours of the target interval. Removing them dropped the figure to 51.7 %; the corrected alignment leaves the served model at 55.2 %. Bitcoin, whose bar closes after the US close, was never affected — its numbers are identical before and after.
- **Deep models.** GRU and LSTM are the weakest family on every asset in validation and not better than the random walk in test (early stopping after 1–12 epochs), even with 3× the earlier data.
- **Stacked ensemble.** With 3 years of data the non-negative meta-weights collapsed to zero; with the full history they are non-zero (Bitcoin: LightGBM 0.39, CatBoost 0.31; Gold: CatBoost 1.18, GRU 0.59; Silver: GRU 0.99, LightGBM 0.55) but the stack does not beat the best single model on test (for Gold it is the worst model there): the base models carry no combinable out-of-sample signal.
- **Over-fitting.** Validation RMSE is 0.8× (Bitcoin), 2.1× (Gold) and 2.9× (Silver) the training RMSE — and the random walk shows the same ratios (Figure 6): the gap is regime shift, not memorisation. The one model that visibly over-fits is Bitcoin's LightGBM (training RMSE 0.0274 against 0.0335 for the random walk, with no validation gain).

9.5 Performance across market regimes (test period, served model vs naive)

Asset	Regime	Days	MAE % served	MAE % naive	Direction hits %
Bitcoin	high volatility	69	1.65	1.66	47.8
Bitcoin	low volatility	69	1.30	1.29	46.4
Bitcoin	medium volatility	69	1.64	1.62	46.4
Bitcoin	down-trend (20d)	96	1.64	1.62	40.6
Bitcoin	up-trend (20d)	111	1.44	1.44	52.3
Gold	high volatility	48	1.33	1.35	52.1
Gold	low volatility	48	1.28	1.27	52.1
Gold	medium volatility	47	1.28	1.26	51.1
Gold	down-trend (20d)	86	1.35	1.35	52.3
Gold	up-trend (20d)	57	1.21	1.21	50.9
Silver	high volatility	48	2.93	2.93	60.4
Silver	low volatility	48	2.11	2.14	60.4
Silver	medium volatility	47	2.40	2.38	44.7
Silver	down-trend (20d)	81	2.57	2.58	59.3
Silver	up-trend (20d)	62	2.36	2.36	50.0

Table 13. Regimes are defined on the day the prediction is made (30-day volatility tercile; sign of the 20-day return). Errors scale with volatility for every asset; no served model improves on the naive forecast in any slice (results/regime_analysis.csv).

9.6 Before vs after the improvement phase (identical unseen days 2026-02-18 → 2026-07-28)

The archived 3-year system and the improved system are compared on exactly the same days. Yahoo Finance revised a few historical closes between the two downloads (mean < 0.005 %, max 0.75 % on one Bitcoin day), so each system is also shown against the naive forecast computed on its own data (results/experiments/before_after.csv).

Asset	System	Model	MAE (\$)	RMSE (\$)	MAPE %	RMSE (ret)	R ² (ret)	Dir. acc. % (p)	DM p
Bitcoin	before (3 y data)	CatBoost	1,096.70	1,424.81	1.59	0.02068	-0.005	47.8 (0.74)	0.57
Bitcoin	after (full history)	CatBoost	1,100.64	1,425.57	1.59	0.02071	-0.007	46.6 (0.83)	0.34
Bitcoin	naive (new data)	Naive	1,092.36	1,420.93	1.58	0.02064	-0.000	—	—
Gold	before (3 y data)	CatBoost	61.74	80.03	1.36	0.01752	-0.041	47.7 (0.72)	0.15
Gold	after (full history)	CatBoost	61.07	79.54	1.35	0.01742	-0.030	49.5 (0.58)	0.62
Gold	naive (new data)	Naive	60.87	78.79	1.34	0.01725	-0.010	—	—
Silver	before (3 y data)	CatBoost	1.97	2.59	2.67	0.03418	-0.017	48.6 (0.65)	0.27
Silver	after (full history)	GRU	1.95	2.57	2.65	0.03403	-0.007	53.2 (0.28)	0.74
Silver	naive (new data)	Naive	1.96	2.57	2.66	0.03399	-0.004	—	—

Table 14. Gold -0.6 %, Silver -0.4 %, Bitcoin +0.1 % return-RMSE — all inside noise. The served models now sit on the random-walk line instead of 0.6–1.5 % below it; the intermediate claim of a 2.8 % Silver improvement was the close-time leak. This comparison reads the test rows a second time; it compares two frozen systems and selects nothing.

9.6b The close-time leak found by the final audit

Yahoo's daily close for GC=F and SI=F is the 13:25–13:30 ET COMEX *settlement*, not the 17:00 ET end of the Globex session. This was verified against 5-minute bars of the December-2026 contracts: the mean absolute difference between the daily close and the 13:30 price is 0.04 % (gold) / 0.09 % (silver), against 0.3–1 % for the 17:00 price. The intermediate version of this system used the *same-day* S&P 500, VIX, 10-year-yield, DXY and WTI returns — fixed between 14:30 and 17:00 ET — and the session High/Low as day-*t* features for the metals. Those values contain up to 3.5 hours of the interval the target measures.

Silver, same 143 unseen days	Model	Dir. acc. (p)	DM p vs naive	R ² (return)
same-day macro features (archived, leak)	LightGBM	62.2 % (0.002)	0.002	+0.044
the five macro features removed	LightGBM	51.7 % (0.37)	0.21	+0.015
previous-day macro + lagged High/Low (final)	GRU (served)	55.2 % (0.12)	0.99	-0.001

Table 14b. The Silver "edge" was the leak (results/archive_sameday_macro_leak/). The effect was invisible in walk-forward validation (dropping the macro group changed CV RMSE by -0.04 %) and only surfaced in a test window where the metals' contemporaneous correlation with equities and rates doubled — a leak that is small on average can dominate one volatile window.

The fix (config.EXTERNAL_SAME_DAY, DataCleaner._align_external, lag_post_settlement_features) uses, for commodity assets, the latest external value *strictly before* day t and the previous session's High/Low-based features; Bitcoin keeps same-day values because its bar closes at 00:00 UTC, after every US close (and its results are bit-identical before and after the fix). Two unit tests enforce the rule. The lesson is stated as a finding of the project: "backward-looking by date" is not the same as "known when the target starts", and a leak of a few hours is enough to manufacture a significant result on daily data.

9.7 Design experiments (validation only)

- **E1 data size** (same features and parameters, fixed validation window 2025-09 → 2026-02): training on 2018 → instead of 2023 → Bitcoin CatBoost -1.1 %; Gold CatBoost -0.2 %; Silver CatBoost -1.4 % validation RMSE — Bitcoin benefits, the metals are neutral; kept because it also stabilises tuning.
- **E2 feature-group ablation** (walk-forward CV, Ridge and LightGBM): dropping any one of the five groups changes RMSE by -0.39 % ... +0.05 % — inside noise; the full set is kept for interpretability.
- **E3 horizon**: a 5-day return target is not more predictable than the next day (best model -1.04 % vs naive); the served horizon stays 1 day.
- **E4 volatility**: 22-day realised volatility is more predictable than the return (Bitcoin Ridge -11.5 % vs persistence) but only marginally better than a RiskMetrics EWMA (-7.7 %) and worse than EWMA for Silver — documented, not productised.

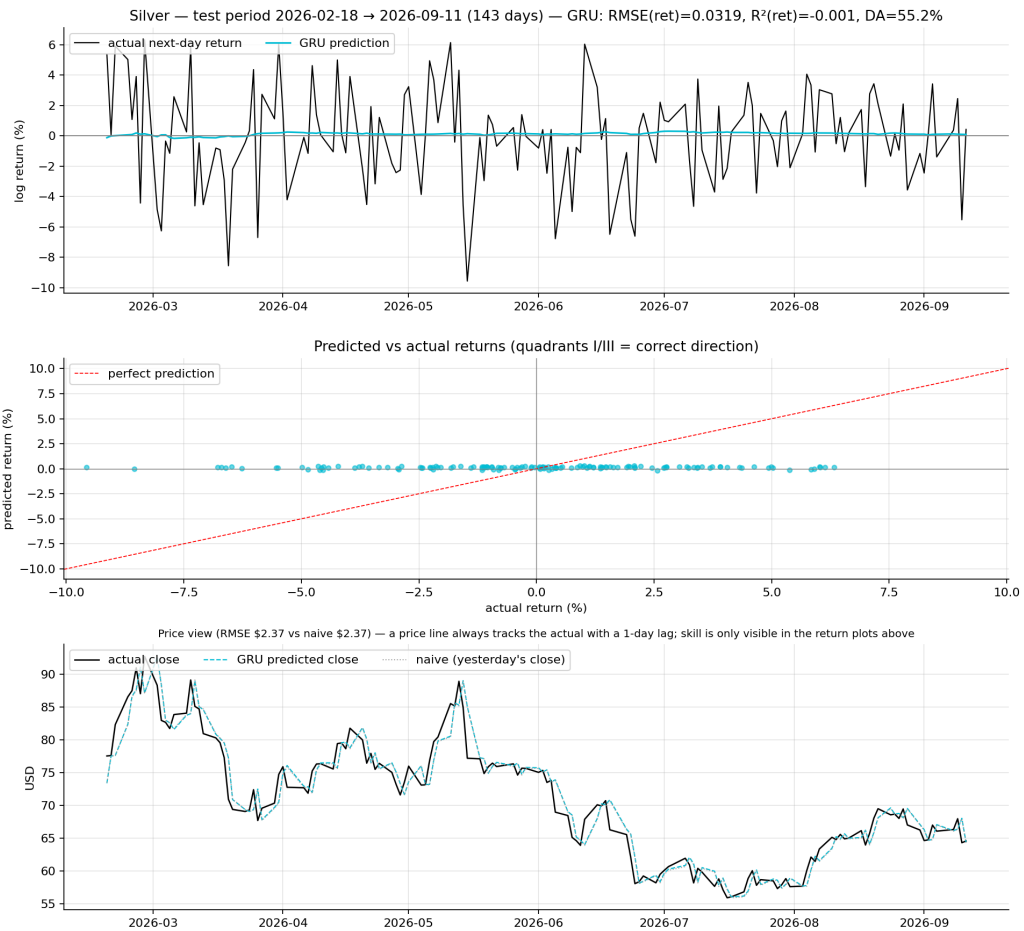


Figure 5. Silver test period, served GRU: predicted vs actual next-day returns (top, scatter middle) and the price view (bottom). A price line always tracks the actual with a one-day lag; skill — or, here, its absence — is only visible in return space.

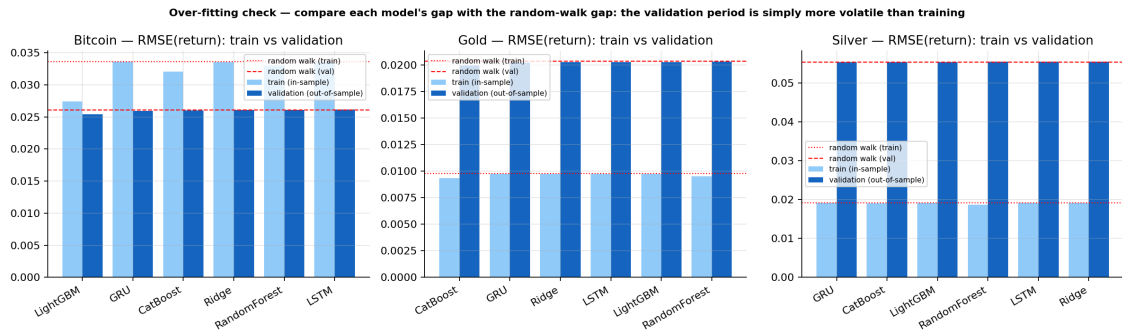


Figure 6. Train vs validation RMSE per model with the random walk's train/validation levels (dotted/dashed).

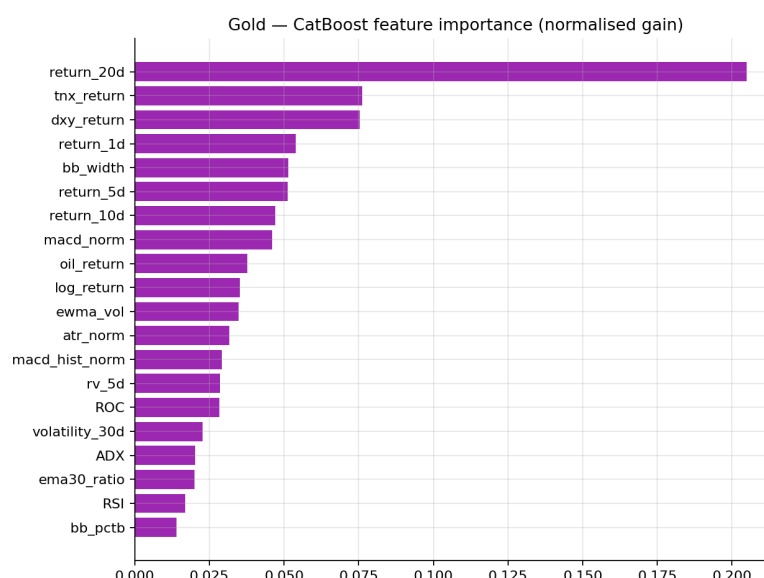


Figure 7. Gold, served CatBoost: normalised feature importance (the served Silver GRU exposes none).

10. Final System and Application

The Streamlit dashboard (make serve, port 8501) and the FastAPI service (make api, port 8000) sit on the same inference module. Inference scales the features up to the chosen day with the train-fitted scaler, runs the requested model on the last 30 rows, un-standardises the predicted return and converts it to a price. The *Predict a Day* tab is the demonstration: stand on any day of the unseen test period, generate the next-day forecast from data up to that day only, then reveal the actual close, the error in dollars and percent, and the direction hit — highlighted on the actual-vs-predicted chart, with every other model's forecast for the same day. The *Prediction History* tab lists every unseen day with predicted, actual, error and hit/miss, plus MAE and hit-rate KPIs. A "Sync Live Market Data" button refreshes features into a separate live file so the frozen evaluation data is never overwritten.

Implemented (verified running)	Not implemented (future possibilities)
• Select Bitcoin / Gold / Silver; price chart with train/val/test shading; Bollinger, RSI, MACD, volume overlays • Predict a Day (unseen test): choose any test day → predicted close, actual close, error \$ and %, direction hit/miss, chart highlight, all models on that day • Prediction History: every unseen day with predicted / actual / error / hit, MAE and hit-rate KPIs, CSV download • Run a next-day prediction with the served model or any trained model: current close, predicted close, % change, direction, horizon (T+1), model name, ±1 RMSE uncertainty band, held-out metrics of that model, disclaimer • Forecast-horizon chart (last 60 days + next day); table of all trained models' predictions on the same input • Validation view: actual vs predicted returns and prices over the test period; return scatter • Performance tab: walk-forward and test tables, honest-reading summary, regime table, experiment tables (E1–E4), report figures • Methodology tab: task, split, features, models, served-model parameters, model inventory • API: GET /health, GET /models, GET /predict/{asset}[?model=...]; Docker Compose for both services • Deep links: ?asset=Gold&run=1	• Multi-day horizons or volatility forecasts • Calibrated prediction intervals (the band is an ex-post error band) • User accounts, saved forecasts, database • Automatic scheduled retraining in production • News / NLP sentiment for the metals • Intraday data • Trading execution or portfolio management

Table 15. Application functionality.

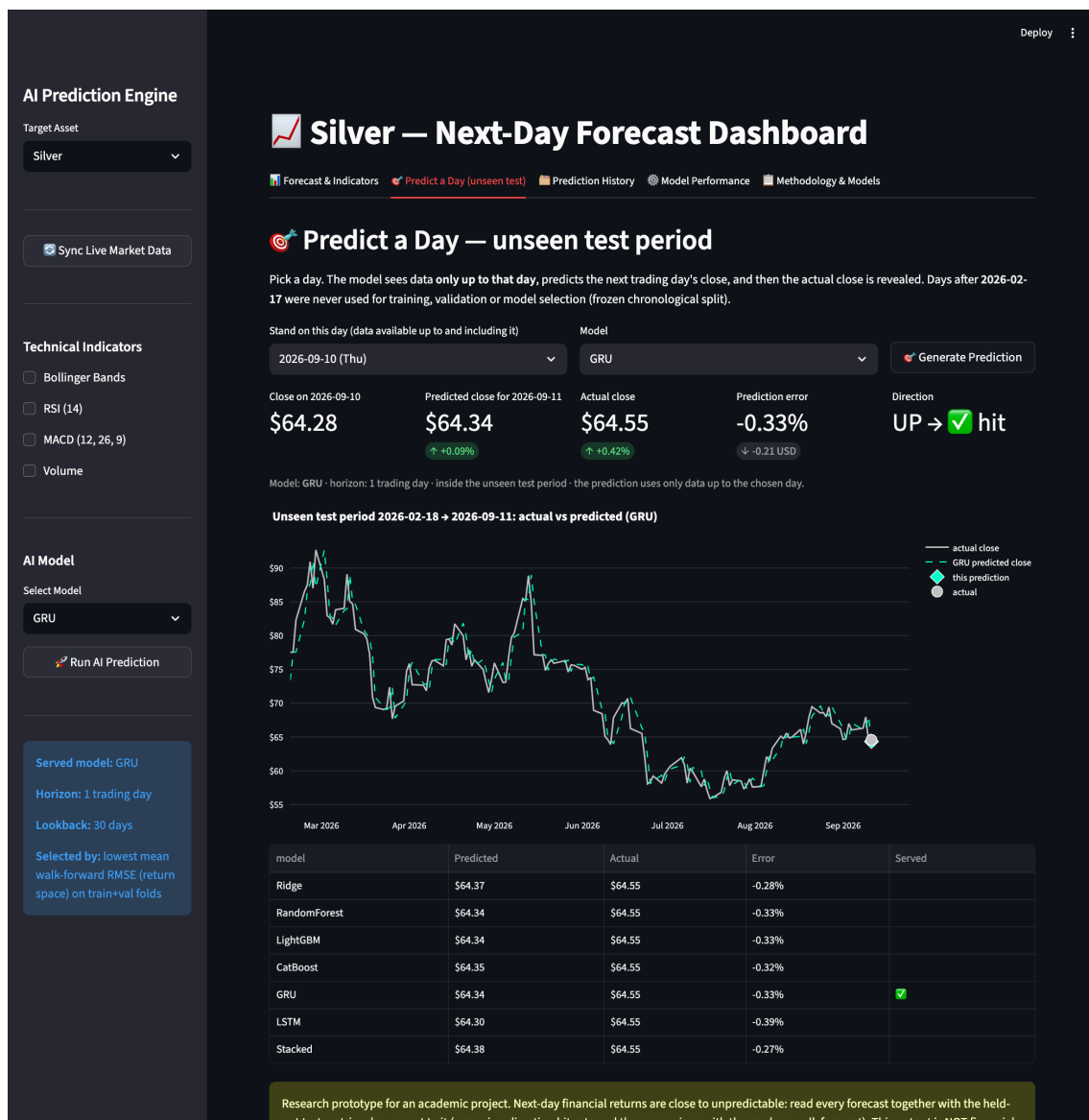


Figure 8. Dashboard, Silver — Predict a Day: the served GRU stands on 2026-09-10, predicts \$64.34 for 2026-09-11 from data up to that day, the actual close (\$64.55) is revealed, error -0.33 %, direction hit; all seven models on the same day below, within \$0.08 of each other (screenshot of the running application).

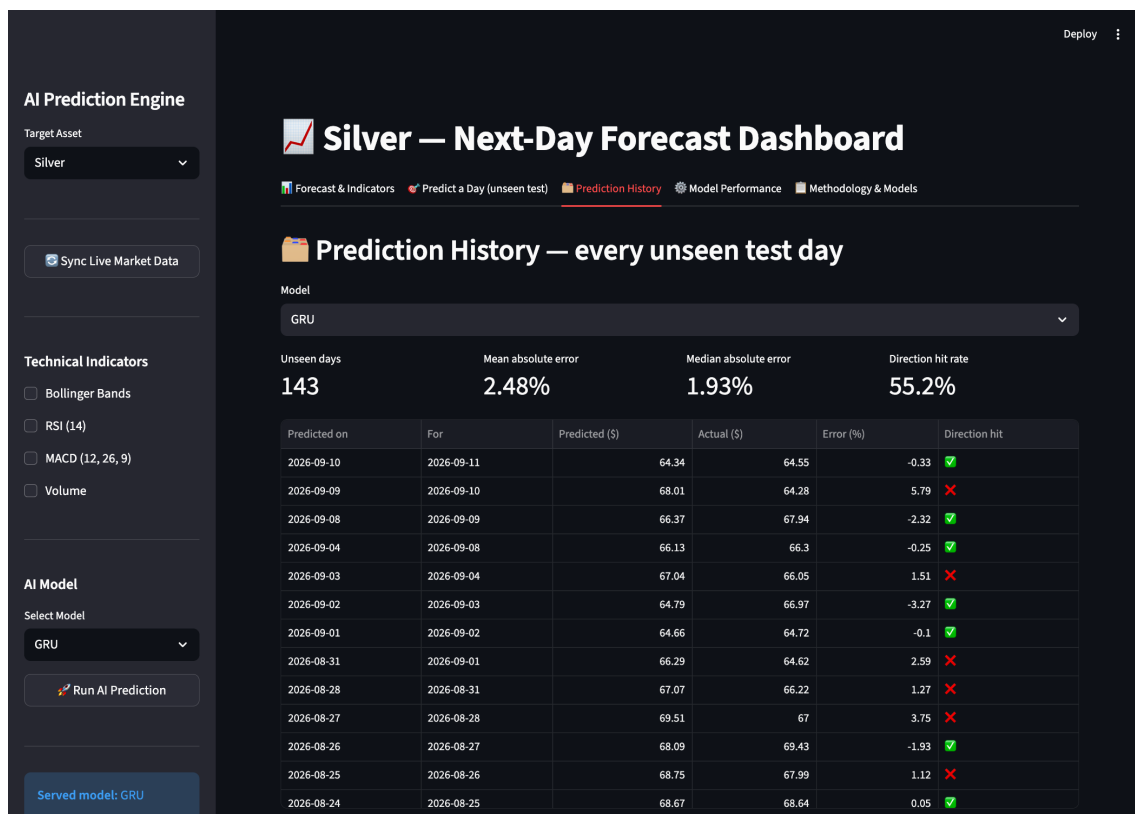


Figure 9. Dashboard, Silver — Prediction History: every unseen day with predicted, actual, error and hit/miss; MAE and hit-rate KPIs.

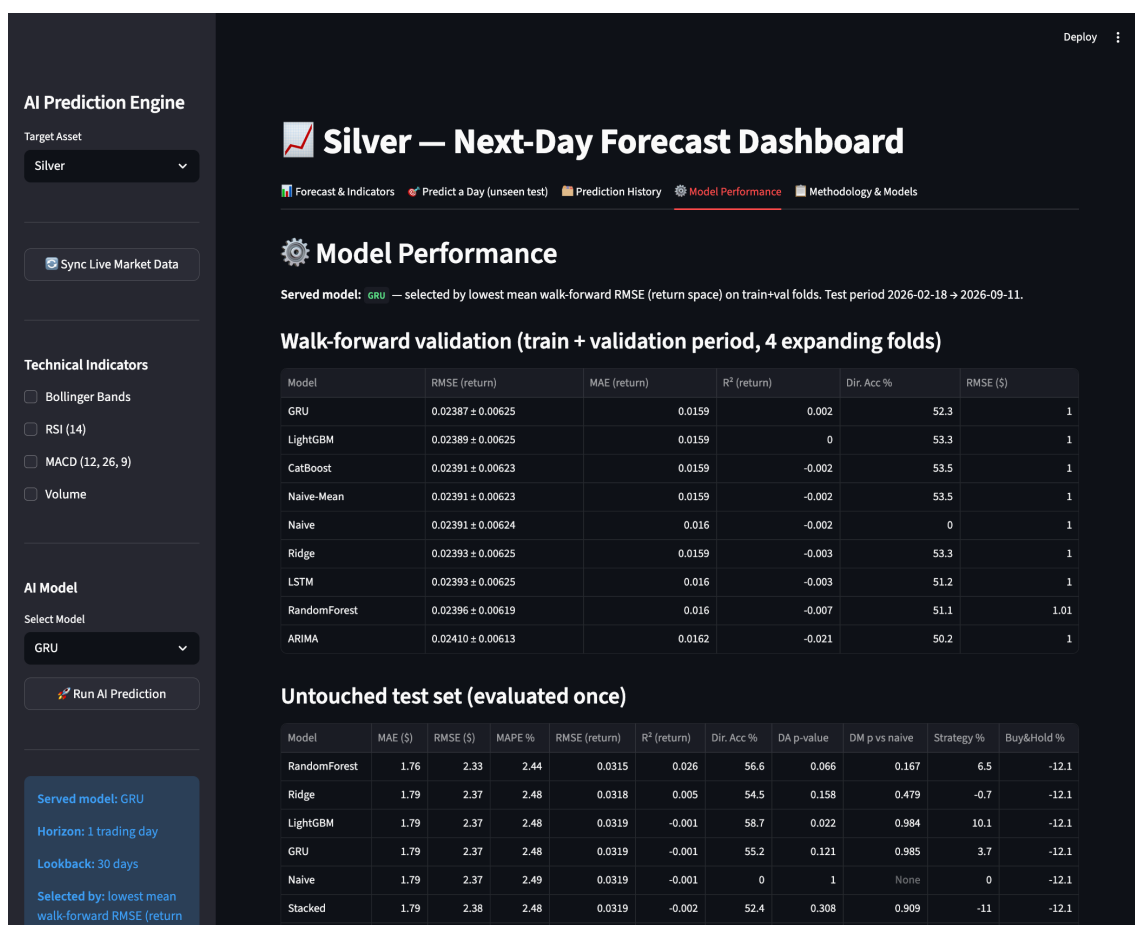


Figure 10. Dashboard — Model Performance tab with the walk-forward and test tables generated by the evaluation script.

11. Improvements Made During the Audit

The initial project already generated predictions, used a log-return target, a chronological split and a train-only scaler, and had a modular code base with a dashboard. The audit nevertheless found problems that invalidated its reported results. All were fixed; the original artefacts are archived under `results/archive_pre_fix/` for the discussion.

Area	Original problem	Final implementation
Price reconstruction	Predicted price rebuilt from the previous day's open (column 0) instead of the close — every USD metric and every dashboard forecast wrong ($\approx \$1,416$ RMSE for Bitcoin from the bug alone)	Explicit previous close; regression test asserts exact reconstruction of the true target
Data pipeline	Gold/Silver forward-filled to calendar days: 30.7 % fabricated zero-return rows, 47 of 159 test days fake	Exchange calendar kept for futures; macro data aligned onto it
Data leakage	Hyper-parameters chosen on the test set (April experiments); CatBoost early-stopped on data inside its training set	Walk-forward tuner on train+val only; two-phase fitting; checklist enforced by tests
Features	Raw price levels (open/high/low/close/EMA/BB/VWAP) as inputs — test values 1.7–2.4× outside the training range	Stationary ratio features; redundant indicators removed; gold return added for silver
Model	Served "stacked ensemble" had coefficients ≈ 0 (Bitcoin exactly [0,0,0]) — a constant predictor, never evaluated on test; 2×100-unit GRU/LSTM over-fitted	Uniform registry of 10 models; validation-based selection; small tunable GRU/LSTM; stack kept as a reported experiment
Evaluation	$R^2 \approx 0.94$ – 0.97 on price levels presented as skill; naive baseline beat every model but was not acknowledged; directional accuracy mis-computed (naive 0 %)	Return-space metrics; correct directional accuracy with p-values; Diebold–Mariano test; strategy backtest; over-fitting gap; honest reporting
Serving	CLI/API crashed (LightGBM given 900 features; float(dict)); dashboard listed stale LSTM and a non-existent Ensemble path	Rewritten inference, API and dashboard around the served model with context and disclaimer; verified in the browser
Code & docs	Five stale result files from an earlier pipeline; Makefile/CI referenced missing scripts; 4 of 31 tests failed; README claimed models/indicators that did not exist	Single results/ tree; working Makefile/CI/retrain; 31 passing tests; accurate README and docs/; bit-identical reproducibility
Improvement phase	3 years of data; no measured design decisions; forecast shown without a way to verify it	2018 → data (2.9×); HAR/EWMA volatility and momentum features; per-task tuning; logged experiments E1–E4; before/after on identical days; Predict-a-Day demo and prediction history
Final audit	Same-day macro and session High/Low features for the metals against a 13:30 ET settlement close → a spurious 62 % Silver result; live sync overwrote frozen raw files; make targets missing a path bootstrap	Close-time rule (previous-day macro, lagged High/Low for metals) with unit tests; pipeline re-run; contaminated results archived; live downloads isolated in <code>data/raw_live/</code> ; reproducible make targets

Table 16. Before vs final implementation (`docs/FIX_PLAN.md`, `docs/AUDIT_SUMMARY.md`, `docs/RESULTS.md` §2–3).

12. Limitations

- **Market efficiency dominates.** Lag-1 autocorrelation of returns is -0.08 / -0.04 / $+0.00$ (BTC / Gold / Silver); at a one-day horizon the predictable component is small and unstable, and the results are consistent with that.
- **One test window, many comparisons.** 8.6 years of training data removed the worst of the small-sample problem, but the unseen window is still a single 5–7-month period (143–207 days): a few percent in RMSE and directional accuracies of 52–57 % are not statistically detectable, and with 27 model-rows in the test table one cell at $p \approx 0.02$ is expected by chance.

- **Regime shift in the test window.** The 2026 window is far more volatile than the 2018–2025 average (validation RMSE 2–3× training for the metals, for the random walk as well as for the models); one window is a single draw from a non-stationary process.
- **Data quality.** Front-month futures (roll effects; unusable volume), a composite BTC index, forward-filled macro series on holidays, a proprietary sentiment index, and vendor revisions of historical closes between downloads (max 0.75 %).
- **Modelling.** One-day point forecasts only; the dashboard's band is an ex-post RMSE band, not a calibrated interval; small tuning grids; selection by RMSE (directional accuracy and Sharpe are reported but not optimised).
- **Backtest.** The long/flat strategy uses a flat 10 bps cost with no slippage, shorting, sizing or financing.

13. Future Work

- Roll the test window forward as new months arrive and re-evaluate walk-forward; use the exchange's official settlement series rather than a free vendor's bar.
- Forecast volatility (GARCH-type or quantile targets), where the volatility-regime features clearly carry information, and score calibration.
- Multi-day horizons and probabilistic forecasts; regime-aware or switching models.
- News/NLP sentiment for the metals; intraday data; a realistic transaction-cost model.

14. Conclusion

The project set out to build a working next-day prediction system for Bitcoin, Gold and Silver and to make it scientifically defensible. The final system has a leakage-audited, reproducible pipeline; ten model families compared under expanding-window walk-forward validation on identical days; a single, statistically tested evaluation on an untouched period; an evaluator-facing demonstration that predicts any unseen day and reveals the actual value and the error; and a dashboard and API that serve the validation-selected model together with its measured error. The empirical answer with 8.6 years of data is unambiguous and honest: for Bitcoin, Gold and Silver no statistical, tree-based or recurrent model beats the random walk at a one-day horizon (served models 0.0–0.6 % above the naive RMSE, Diebold–Mariano $p \geq 0.71$, directional accuracy 47–55 %). The 2.9× larger training set moved the models from slightly below the random walk onto it, not above it. The value of the work lies in the rigour of the evaluation — in showing why the original " $R^2 \approx 0.95$ " was an illusion, and in finding and removing a close-time leak in its own intermediate version that had manufactured a "significant" 62 % result — rather than in a claimed forecasting edge.

15. FYP Readiness Assessment

Suitable for submission and live demonstration. The implementation demonstrates the complete ML research workflow (problem → data → analysis → features → baselines → models → validation → experiments → comparison → final model → system → evaluation → limitations); every methodological claim is backed by a test, a script or a results file; the application runs for all three assets with an interactive unseen-day demonstration; every design change was measured before being kept; and the pipeline reproduces every reported number. The remaining weakness is the single test window, which is stated in the limitations. It should be presented as a methodological benchmark and decision-support prototype — including the leak it found in itself — not as a market-beating predictor.

Appendix — Reproduction

```
python -m venv venv && source venv/bin/activate && pip install -r requirements.txt
make preprocess eda tune train stack evaluate figures experiments test # full pipeline from
```

```
data/raw  
make serve # dashboard on http://localhost:8501  
make api # FastAPI on http://localhost:8000/docs  
python docs/build_report.py # regenerate this report from results/
```

Repository documents: docs/METHODOLOGY.md, docs/FEATURES.md, docs/RESULTS.md, docs/LIMITATIONS.md, docs/VIVA_QA.md, docs/DEMO_GUIDE.md, docs/REPORT_STRUCTURE.md, docs/FIX_PLAN.md, docs/AUDIT_SUMMARY.md, results/FINAL_RESULTS.md, results/experiments/experiments_summary.md.